INTEGRATION QUIZ

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Name: Solutions

1. Use right endpoints and the formulas

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

to evaluate the definite integral

$$\int_1^2 x^2 \, \mathrm{d}x$$

$$\Delta x = \frac{2-1}{n} = \frac{1}{n}, \quad x_i = 1 + i\left(\frac{1}{n}\right)$$

$$R_n = \sum_{i=1}^{n} (1+i/n)^2\left(\frac{1}{n}\right) = \sum_{i=1}^{n} \left(1 + 2\left(\frac{i}{n}\right) + \left(\frac{i}{n}\right)^2\right)\left(\frac{1}{n}\right) = \sum_{i=1}^{n} \left(\frac{1}{n} + \frac{2}{n^2} i + \frac{1}{n^3} i^2\right)$$

$$= \sum_{i=1}^{n} \frac{1}{n} + \sum_{i=1}^{n} \frac{2}{n^2} i + \sum_{i=1}^{n} \frac{1}{n^3} i^2 = \frac{1}{n}\sum_{i=1}^{n} 1 + \frac{2}{n^2}\sum_{i=1}^{n} i + \frac{1}{n^3}\sum_{i=1}^{n} i^2$$

$$= \frac{1}{n}(n) + \frac{2}{n^2}\left(\frac{n(n+1)}{2}\right) + \frac{1}{n^3}\left(\frac{n(n+1)(2n+1)}{6}\right) = 1 + \frac{n+1}{n} + \frac{2n^2+3n+1}{6n^2} = 1 + 1 + \frac{1}{n} + \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}$$

$$= 7/3 + \frac{1}{n} + \frac{1}{2n} + \frac{1}{6n^2}$$

$$\int_1^2 x^2 dx = \lim_{n\to\infty} R_n = \lim_{n\to\infty}\left(7/3 + \frac{1}{n} + \frac{1}{2n} + \frac{1}{6n^2}\right) = \frac{7}{3} + 0 + 0 + 0 = \boxed{\frac{7}{3}}$$